Introduction to

Generative AI

Complete Study Notes

Generative AI · Study Reference

These notes are derived from the IBM/Coursera lecture on Generative AI. They are structured for quick revision and long-term reference.

1. What is Artificial Intelligence?

Artificial Intelligence (AI) is the simulation of human intelligence by machines. AI systems are designed to perform tasks that normally require human cognition — such as recognising patterns, making decisions, understanding language, and generating content.

Training: The process through which an AI model learns from data is called training. During training, a model is exposed to vast amounts of data and adjusts its internal parameters to minimise errors in its predictions or outputs.

2. Two Fundamental Approaches to AI

There are two broad approaches to AI, each suited for different tasks:

Discriminative AI	Generative AI

Discriminative AI learns to distinguish between different classes of data. It is given labelled training data and finds a decision boundary to classify new inputs. **Key abilities:** Classify, differentiate, identify patterns, draw conclusions. **Limitation:** Cannot understand context or generate new content. **Classic example:** An email spam filter that separates spam from legitimate email. **Best for:** Classification tasks — image recognition, fraud detection, sentiment analysis.	Generative AI learns to generate entirely new content based on patterns in training data. It captures the underlying data distribution and produces novel instances. **Key abilities:** Generate text, images, audio, video, code, and data. **Input (prompt):** Text, image, video, or any processable input. **Classic example:** Responding to 'Draw a nest with three eggs in it'. **Best for:** Creative tasks — content creation, code generation, design, synthesis.

Analogy: Discriminative AI mimics our analytical and predictive skills. Generative AI goes a step further — it mimics our creative skills. As the Harvard Business Review notes, AI can not only boost analytic and decision-making abilities but also heighten creativity.

Output modalities: Generative AI can produce output in the same form as the prompt (text-to-text, image-to-image) or in a different form (text-to-image, image-to-video, text-to-audio).

3. Deep Learning — The Foundation

Both discriminative and generative models are built using deep learning techniques.

Deep learning involves training Artificial Neural Networks (ANNs) on large amounts of data. An ANN is a collection of smaller computing units called **neurons**, connected in layers and modelled loosely on how the human brain processes information. Deep learning is so named because these networks have many layers (depth).

4. Building Blocks of Generative AI

The creative capabilities of generative AI come from four core model architectures. These are considered the foundational building blocks:

GANs	**Generative Adversarial Networks** Two networks — a generator and a discriminator — compete against each other. The generator creates fake data; the discriminator judges if it's real or fake. Through this competition, the generator improves until it produces convincing outputs. Introduced by Ian Goodfellow et al. in 2014, GANs were a landmark moment in generative AI.

VAEs	**Variational Autoencoders** Learn a compressed (latent) representation of input data. The encoder maps inputs to a distribution in latent space; the decoder samples from it to reconstruct or generate new data. Useful for image generation, anomaly detection, and data augmentation.
Transformers	**Attention-based Architecture** Use a self-attention mechanism to process sequences of data in parallel rather than sequentially. This enables understanding of long-range dependencies in text. The backbone of all modern LLMs including GPT, PaLM, and BERT. Introduced in the paper 'Attention Is All You Need' (Vaswani et al., 2017).
Diffusion Models	**Denoising Process** Learn to reverse a gradual noising process applied to training data. During training, noise is added to images step-by-step; the model learns to remove it. At inference, it starts from pure noise and iteratively denoises to produce a clean image. Powers Stable Diffusion and DALL-E image generators.

5. Evolution of Generative AI — Timeline

Generative AI is not new — its roots trace back to the very beginnings of machine learning:

1950s	Scientists proposed machine learning and began exploring algorithms that could create new data from learned patterns.
1990s	The rise of neural networks infused significant advancements in generative AI capabilities.
Early 2010s	Deep learning, powered by large datasets and enhanced computing hardware (GPUs), dramatically advanced generative AI.
2014	Ian Goodfellow and colleagues introduced Generative Adversarial Networks (GANs) — a transformative moment that opened new possibilities for content generation.
2017	Google researchers published 'Attention Is All You Need', introducing the Transformer architecture that underpins all modern LLMs.
2018	OpenAI released GPT (Generative Pre-trained Transformer), a transformer-based large language model trained on internet-scale text data.
2020–2022	GPT-3, DALL-E, Stable Diffusion, and other powerful models launched — bringing generative AI to the mainstream.
2023–Present	GPT-4, Gemini, LLaMA, Claude, Midjourney V5+ and many others significantly raise the bar for quality and capability across all modalities.

6. Foundation Models and LLMs

Foundation Models

Foundation models are large AI models trained on broad, diverse data that give them general capabilities. They serve as a flexible base that can be fine-tuned or adapted to create more specialised models or tools for specific use cases — without training from scratch.

Think of a foundation model like a highly educated generalist. Fine-tuning is like giving them specialised on-the-job training to become an expert in a specific domain (e.g. medicine, law, coding).

Large Language Models (LLMs)

LLMs are a specific category of foundation models trained on massive text datasets. They learn statistical relationships between words, phrases, and concepts, enabling them to:

- Understand and generate human language with high fluency
- Answer questions, summarise documents, write code, translate languages
- Engage in multi-turn conversations and follow complex instructions
- Perform few-shot and zero-shot learning — generalising from minimal examples

Notable LLMs:

Model	Developer	Key Feature
GPT-3 / GPT-4	OpenAI	Industry-leading text generation; powers ChatGPT
PaLM / Gemini	Google	Pathways architecture; multimodal capabilities
LLaMA	Meta	Open-weights model; widely used for research
Claude	Anthropic	Safety-focused; long context window

Image Generation Models

Similar progress has been made for image generation. Key models include:

- **DALL-E / DALL-E 2** (OpenAI) — Text-to-image generation using a combination of CLIP and diffusion.
- **Stable Diffusion** (Stability AI) — Open-source latent diffusion model widely used for creative image generation.
- **MidJourney** — Highly capable text-to-image model known for artistic and photorealistic outputs.

7. Popular Generative AI Tools by Use Case

The rapid development of generative models has created a growing ecosystem of tools for diverse use cases:

Text Generation	Image Generation	Video Generation	Code Generation
ChatGPT Gemini Claude Copilot Chat	DALL-E 2 MidJourney Stable Diffusion Adobe Firefly	Synthesia Runway ML Sora (OpenAI) Pika	GitHub Copilot AlphaCode Tabnine CodeWhisperer

8. Applications Across Domains

Generative AI has a wide scope for transformative applications across nearly every industry:

Healthcare	Drug discovery, medical report generation, radiology image synthesis, personalised treatment plans.

Education	Personalised tutoring, automated grading, content generation for courses, language learning.
Marketing & Media	Ad copy generation, image/video creation, personalised content at scale, A/B test variants.
Software Development	Code autocompletion, bug detection, automated documentation, test generation.
Finance	Report drafting, risk analysis narratives, fraud pattern synthesis, customer service chatbots.
Creative Arts	Music composition, screenplay writing, graphic design, game asset generation.

McKinsey Report: Generative AI has the potential to change the anatomy of work, augmenting the capabilities of individual workers by automating some of their individual activities. Its impact on productivity could add trillions of dollars in value to the global economy.

9. Key Takeaways

1. Generative AI models learn to generate new content — text, images, audio, video, code — based on patterns in their training data.
2. The creative power of generative AI comes from four core architectures: GANs, VAEs, Transformers, and Diffusion Models.
3. Foundation models are broad, adaptable AI models that can be fine-tuned into specialised tools for specific domains.
4. LLMs (Large Language Models) are a subset of foundation models trained specifically on text — they power tools like ChatGPT, Gemini, and Claude.
5. Generative AI differs from discriminative AI: discriminative AI classifies and predicts; generative AI creates.
6. The field has evolved rapidly — from early ML experiments in the 1950s to today's multimodal billion-parameter models.
7. McKinsey predicts generative AI's impact on productivity could add trillions of dollars to the global economy.